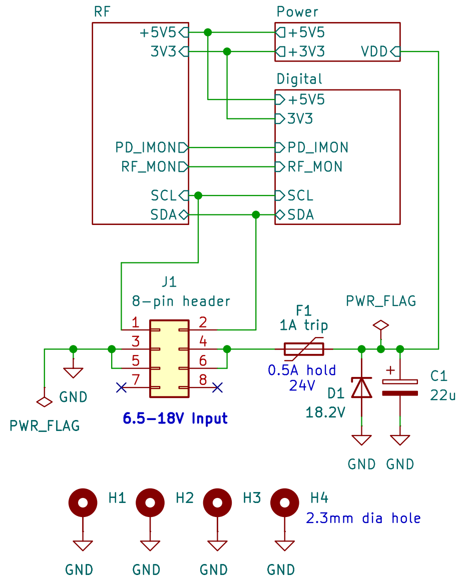
I2C Adresses:

Attenuator	0x20
ADC	0x10
TEMP & ID	0x48



Requires 3/8in 2-56 standoffs for board-to-board spacing.

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Sheet: /
File: frx.kicad_sch

Title: FRX schematic

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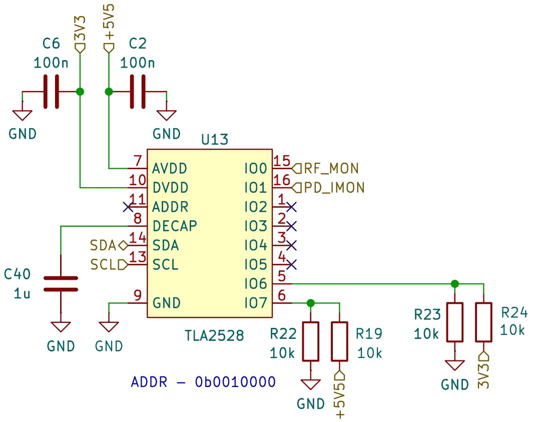
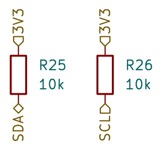
KiCad E.D.A. 8.0.4		Id: 1/4

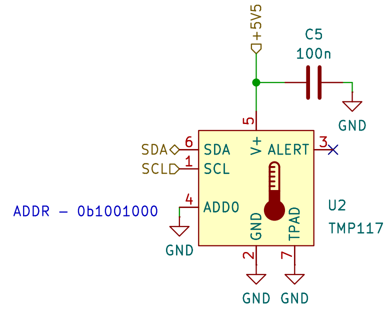
ADC & digital I/O





Weak I2C Pullups
Technically, the pullups should be on the controller,
so these are weak to support that case.

Temp sensor & ID



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Title: FRX schematic

Size: A4	Date: 2024-01-19	**Rev: 2**
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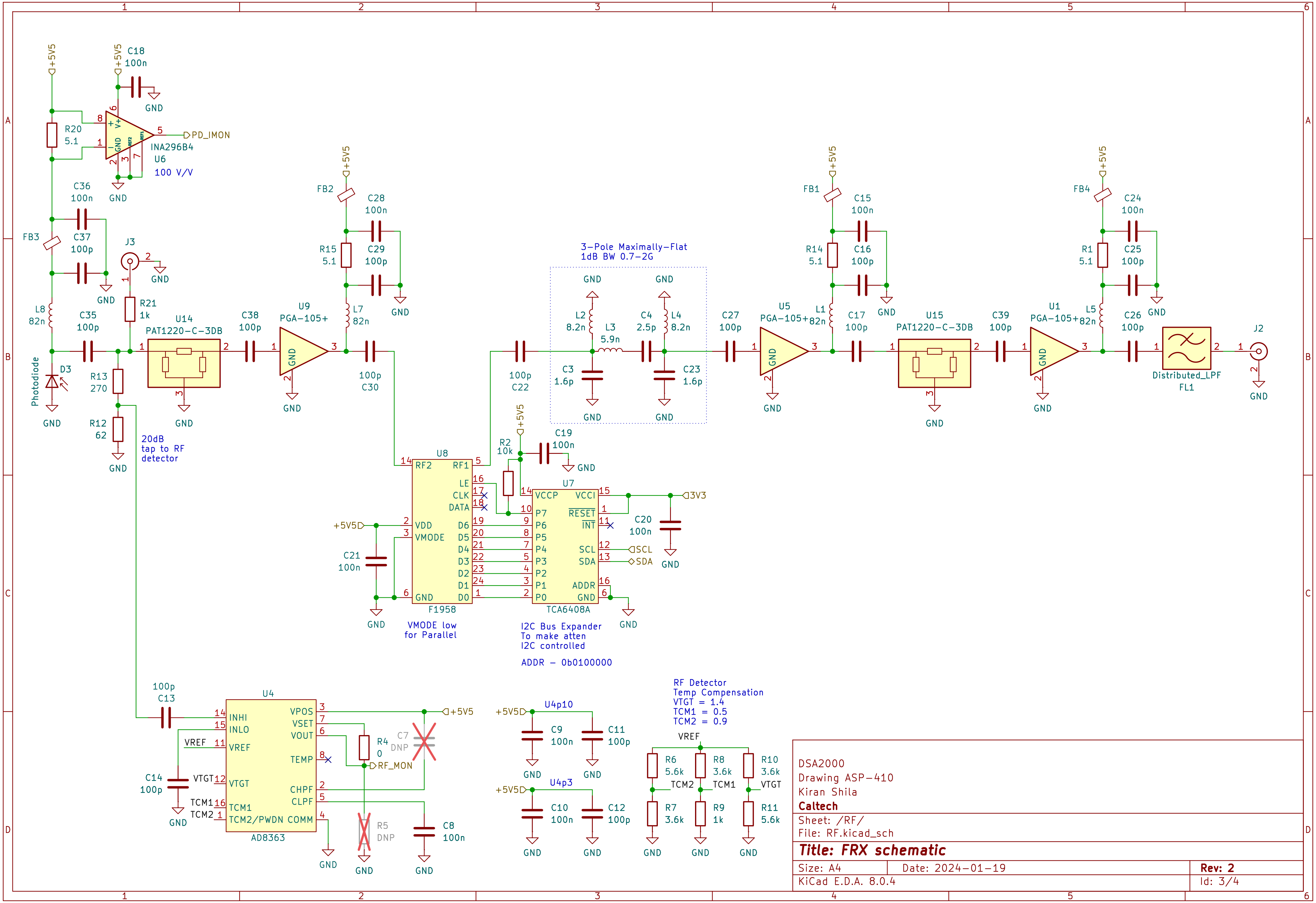

+5V5
C18
100n
GND
R20
5.1
INA296B4
U6
100 V/V
PD_IMON
C36
100n
C37
100p
FB3
J3
R21
1k
L8
82n
C35
100p
U14
PAT1220-C-3DB
C38
100p
U9
PGA-105+
FB2
C28
100n
R15
5.1
C29
100p
L7
82n
100p
C30
Photodiode
D3
R13
270
R12
62
20dB
tap to RF
detector
3-Pole Maximally-Flat
1dB BW 0.7-2G
L2
8.2n
L3
5.9n
C4
2.5p
L4
8.2n
C3
1.6p
C23
1.6p
100p
C22
C27
100p
U5
PGA-105+
FB1
C15
100n
R14
5.1
C16
100p
L1
82n
C17
100p
U15
PAT1220-C-3DB
C39
100p
U1
PGA-105+
FB4
C24
100n
R1
5.1
C25
100p
L5
82n
C26
100p
Distributed_LPF
FL1
J2
R2
10k
C19
100n
U8
RF2
RF1
LE
CLK
DATA
VDD
VMODE
D6
D5
D4
D3
D2
D1
GND
D0
F1958
C21
100n
VMODE low
for Parallel
U7
VCCP
VCCI
P7
RESET
INT
P6
P5
P4
P3
P2
P1
P0
SCL
SDA
ADDR
TCA6408A
3V3
C20
100n
I2C Bus Expander
To make atten
I2C controlled
ADDR - 0b0100000
100p
C13
U4
INHI
INLO
VREF
VTGT
TCM1
TCM2/PWDN
VPOS
VSET
VOUT
TEMP
CHPF
CLPF
COMM
AD8363
C14
100p
R4
0
C7
DNP
RF_MON
R5
DNP
C8
100n
U4p10
C9
100n
C11
100p
U4p3
C10
100n
C12
100p
RF Detector
Temp Compensation
VTGT = 1.4
TCM1 = 0.5
TCM2 = 0.9
VREF
R6
5.6k
R8
3.6k
R10
3.6k
R7
3.6k
R9
1k
R11
5.6k
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Title: FRX schematic
Size: A4
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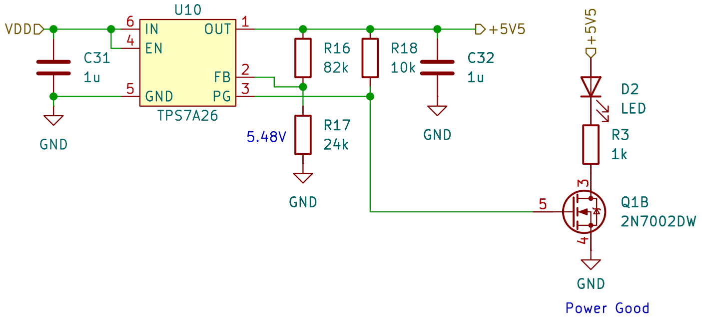
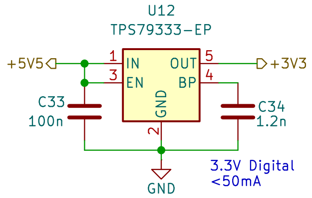
+5.5V Regulated Supply



+3.3V Digital Supply



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Sheet: /Power/
File: power.kicad_sch

Title: FRX schematic

Size: A4	Date: 2024-01-19	**Rev: 2**
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